Cambridge Assessment International Education

# Cambridge O Level

CANDIDATE NAME

CENTRE NUMBER

CANDIDATE NUMBER

*9517397878*

**COMPUTER SCIENCE** **2210/12**

Paper 1 Theory **May/June 2022**

**1 hour 45 minutes**

You must answer on the question paper.

No additional materials are needed.

**INSTRUCTIONS**

- Answer **all** questions.
- Use a black or dark blue pen. You may use an HB pencil for any diagrams or graphs.
- Write your name, centre number and candidate number in the boxes at the top of the page.
- Write your answer to each question in the space provided.
- Do **not** use an erasable pen or correction fluid.
- Do **not** write on any bar codes.
- Calculators must **not** be used in this paper.

**INFORMATION**

- The total mark for this paper is 75.
- The number of marks for each question or part question is shown in brackets [ ].
- No marks will be awarded for using brand names of software packages or hardware.

This document has **12** pages.

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**[Turn over**

**1** **(a)** Denary values are converted to binary values to be processed by a computer.

Draw **one** line from each denary value to the correctly converted 8-bit binary value.

| Denary | 8-bit binary |
|---|---|
| | 00100001 |
| 41 | 10100110 |
| | 00101001 |
| 174 | 10000110 |
| | 10101110 |
| 86 | 01010110 |

[3]

Working space

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**(b)** Binary values can also be converted to denary values.

Give the correct denary value for the 12-bit binary value 000101010111
Show all your working.

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Denary value .....

[2]

**2** Hexadecimal is used for Hypertext Markup Language (HTML) colour codes.

An HTML colour code is:

#2F15D6

Each pair of digits is stored as binary in an 8-bit register.

**(a)** Give the 8-bit binary value that would be stored for each pair of hexadecimal digits.

| | | | | | | | | |
|---|---|---|---|---|---|---|---|---|
| 2F | | | | | | | | |
| 15 | | | | | | | | |
| D6 | | | | | | | | |

[6]

Working space

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**(b)** HTML colour codes and Media Access Control (MAC) addresses are two examples of where hexadecimal is used in Computer Science.

Give **two** other examples of where hexadecimal can be used in Computer Science.

Example 1 .....

Example 2 .....

[2]

**(c)** Websites can be created using HTML structure and presentation.

State what is meant by HTML structure and presentation.

Give an example of each in your answer.

Structure .....

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Presentation .....

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[4]

**(d)** Explain why presentation is often separated from structure when creating a web page.

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..... [2]

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**3** Joelle is a student who uses the Internet.

**(a)** The table contains **five** terms or definitions that relate to the Internet.

Complete the table by writing each missing term or definition.

| Term | Definition |
|---|---|
| browser | ..... ..... ..... |
| ..... | this is the company that provides a user with a connection to the Internet |
| ..... | this is a protocol that is used to send data for web pages across the Internet |
| Uniform Resource Locator (URL) | ..... ..... ..... |
| cookie | ..... ..... ..... |

[5]

**(b)** Joelle uses a firewall to keep her data safe when she uses the Internet.

Tick (✓) to show which statement about firewalls is true.

| | **Tick (✓)** |
|---|---|
| Firewalls can only be hardware-based | ☐ |
| Firewalls can only be software-based | ☐ |
| Firewalls can be hardware-based or software-based | ☐ |

[1]

**(c)** Joelle's parent also uses the firewall to limit the websites that Joelle can access.

Explain how the firewall is used to limit the websites that Joelle can access.

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..... [4]

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**4** Jason is a programmer who writes computer programs in a high-level language.

**(a)** Describe what is meant by a high-level language.

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..... [3]

**(b)** Jason wants to distribute a computer program he has written. He is considering distributing it to users as freeware or free software.

**(i)** Explain **one** drawback to a user if the program is distributed as freeware.

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..... [2]

**(ii)** Explain **one** benefit to a user if the program is distributed as free software.

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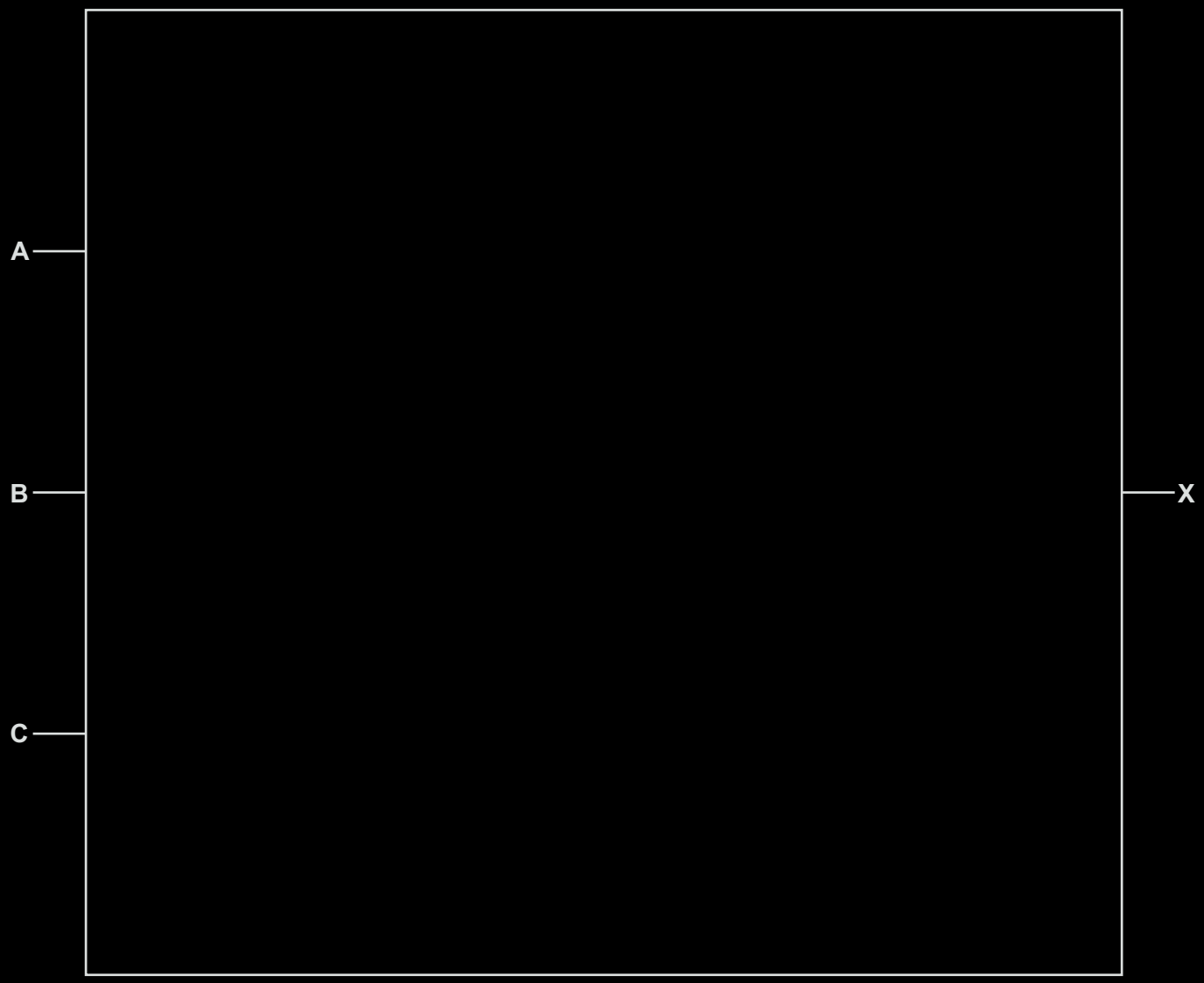
..... [2]

**5** Consider the following logic statement:

**X** = ((**A** OR **B**) AND (NOT (**B** XOR **C**)) AND **C**)

**(a)** Draw a logic circuit to represent the given logic statement.

Do **not** attempt to simplify the logic statement. All logic gates must have a maximum of **two** inputs.




[5]

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**(b)** Complete the truth table for the given logic statement.

| A | B | C | Working space | X |
|---|---|---|---|---|
| 0 | 0 | 0 | | |
| 0 | 0 | 1 | | |
| 0 | 1 | 0 | | |
| 0 | 1 | 1 | | |
| 1 | 0 | 0 | | |
| 1 | 0 | 1 | | |
| 1 | 1 | 0 | | |
| 1 | 1 | 1 | | |

[4]

**6** Millions of emails are sent between users on a daily basis.

**(a)** Identify **two** online security attacks that can be carried out using email.

Describe how email is used to enable the attack.

Online security attack 1 .....

Description .....

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Online security attack 2 .....

Description .....

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[6]

**(b)** Online security attacks can maliciously damage data.

One security method to keep data safe from online attacks is a firewall.

Identify **two** other security methods that keep data safe from online attacks.

Security method 1 .....

Security method 2 .....

[2]

**(c)** Data can also be damaged accidentally.

One example of how data can be damaged accidentally is by shutting down a computer before saving data. To prevent this from happening, a user should make sure they have saved all data before shutting down a computer.

Complete the table by giving **three** other examples of how data can be damaged accidentally.

Give a method of prevention for each example.

| Example | Method of prevention |
|---|---|
| ..... | ..... |
| ..... | ..... |
| ..... | ..... |
| | ..... |
| ..... | ..... |
| ..... | ..... |
| ..... | ..... |
| | ..... |
| ..... | ..... |
| ..... | ..... |
| ..... | ..... |
| | ..... |

[6]

**7** Cassie stores data for her business every day. She stores the data using optical data storage.

**(a)** Identify **three** examples of optical data storage.

Example 1 .....

Example 2 .....

Example 3 .....

[3]

**(b)** **Six** statements are given about the operation of three different types of storage.

Tick (✓) to show which statements apply to each type of storage. Some statements may apply to more than **one** type of storage.

| | Type of storage | | |
|---|---|---|---|
| **Statement** | **Magnetic (✓)** | **Optical (✓)** | **Solid state (✓)** |
| this storage has no moving parts | | | |
| this storage uses a laser to read and write data | | | |
| this storage uses a read/write head | | | |
| this storage burns pits onto a reflective surface | | | |
| this storage uses NAND and NOR technology | | | |
| this storage stores data in tracks and sectors | | | |

[6]

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[Turn over

**8** Sam develops a software application. He distributes a version of the software as shareware.

**(a)** Describe what is meant by shareware.

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..... [4]

**(b)** Identify **three** ethical issues that may need to be considered when developing and distributing software.

Ethical issue 1 .....

Ethical issue 2 .....

Ethical issue 3 .....

[3]

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